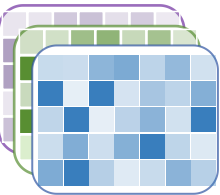


a Input Data & Processing

Paired Count Matrices

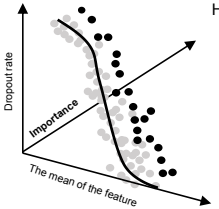


RNA ATAC ADT

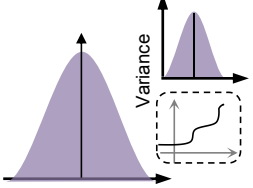
Normalization

	$n_{\text{genes/peaks}}$	...
n_{cells}	0 0.8 0 ...	0 0.9 0.1 ...
0.6	0 0 0 ...	0.1 0 0 ...
0	0 0 0.1 ...	0 0.3 0.9 ...
...	...	...

Highly Variable Genes



Standardization



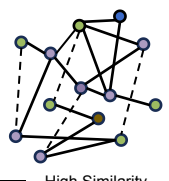
Output: standardized matrix
Integrated multi-omics features used for latent, representation learning and downstream tasks

c Graph-Aware Residual Quantization (GARQ)

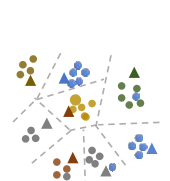
Latent Embeddings



Cosine KNN Graph



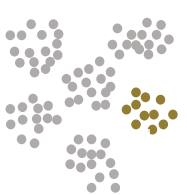
Anchor Assignment



By leveraging local graph structure and residual refinement, GARQ prevents rare cell states from being forced into dominant anchors.

$$Q(h_i) = a_k^*, k^*$$
$$= \arg \max \left\{ \cos(h_i, a_k) + \alpha \sum_{j=1}^n s_{ij} \cos(h_j, a_k) \right\}$$

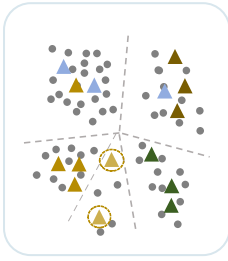
Rare-State Preservation



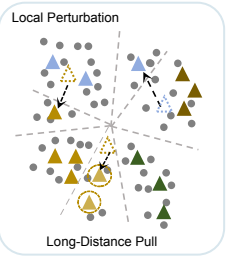
Iterate Until Convergence

d Dynamic Anchor Adjustment (GARQ)

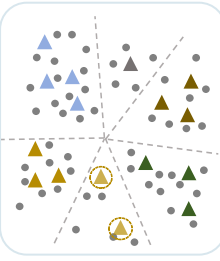
1. Anchor Usage Diagnosis



2. Anchor Adjustment



3. Anchor Balance



EMA-based usage estimates adaptively strengthen relocation for underused anchors and perturbation for overused anchors.

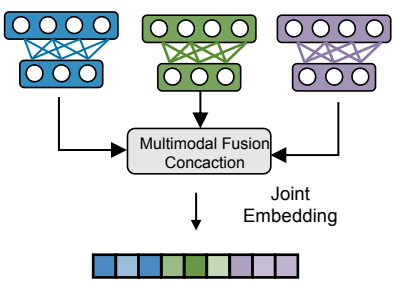
$$\text{EMA: } U_k^t = \eta U_k^{t-1} + (1 - \eta) \frac{n_k^t}{n^t}$$

$$\text{Underused: } \beta_u^t = \exp(-U_k^t \cdot K \cdot 100 - \epsilon)$$

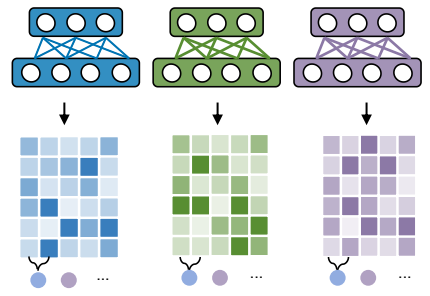
$$\text{Overused: } \beta_o^t = \exp(-10 \cdot \frac{K}{n} \cdot \frac{1}{u_k^t} - \epsilon)$$

---> Move
▲ Anchor (Colored By Cell Types)
● Rare Anchor
● Common Cells (Colored By Cell Types)
● Rare Cells

b Multi-Model Encoding

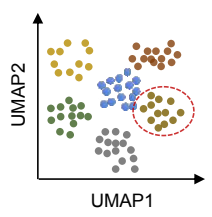


e Multi-Model Decoding



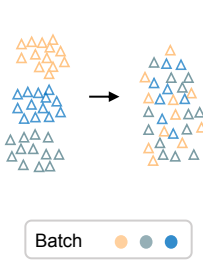
f Application

Visualisation



Rare Cell States

Batch Correction



Biological Analysis

